T1

$$\text{minimize} \quad z = 3x_1 + 2x_2 + 4x_3$$
$$\text{s.t.} \quad 2x_1 + x_2 + 3x_3 = 60$$
$$3x_1 + 3x_2 + 5x_3 \geq 120$$
$$x_1, x_2, x_3 \geq 0$$

introduce the artificial variable $\bar{x}_4$, $\bar{x}_6$.
slack variable x_5

So the original LP problem can be transformed into

$$\text{maximize} \quad z = -3x_1 - 2x_2 - 4x_3 - M(\bar{x}_4 + \bar{x}_6)$$
$$\text{s.t.} \quad 2x_1 + x_2 + 3x_3 + \bar{x}_4 = 60$$
$$3x_1 + 3x_2 + 5x_3 - x_5 + \bar{x}_6 = 120$$
$$x_1, x_2, x_3, \bar{x}_4, x_5, \bar{x}_6 \geq 0.$$

$$\because \begin{cases} \bar{x}_4 = 60 - 2x_1 - x_2 - 3x_3 \\ \bar{x}_6 = 120 - 3x_1 - 3x_2 - 5x_3 + x_5 \end{cases}$$

$$\therefore z = -3x_1 - 2x_2 - 4x_3 - M(60 - 2x_1 - x_2 - 3x_3 + 120 - 3x_1 - 3x_2 - 5x_3 + x_5)$$

$$= -180M + (5M-3)x_1 + (4M-2)x_2 + (8M-4)x_3 - Mx_5$$

$$\Rightarrow z + (3-5M)x_1 + (2-4M)x_2 + (4-8M)x_3 + Mx_5 = -180M.$$

Transform into the form:

Iteration	BV	row	Coeifficient							RHS	Ratio
			z	x_1	x_2	x_3	$\bar{x}_4$	x_5	$\bar{x}_6$		
0	z	(0)	1	$3-5M$	$2-4M$	$4-8M$	0	M	0	$-180M$	
	$\bar{x}_4$	(1)	0	2	1	3	1	0	0	60	20
	$\bar{x}_6$	(2)	0	3	3	5	0	-1	1	120	24

1	z	(0)	1	$\frac{1}{3}M+\frac{1}{3}$	$\frac{2}{3}-\frac{4}{3}M$	0	$\frac{8M-4}{3}$	M	0	$-20M-80$	
	x_3	(1)	0	$\frac{2}{3}$	$\frac{1}{3}$	1	$\frac{1}{3}$	0	0	20	60
	$\bar{x}_6$	(2)	0	$-\frac{1}{3}$	$\frac{4}{3}$	0	$-\frac{5}{3}$	-1	1	20	15
2	z	(0)	1	$\frac{3}{2}$	0	0	$3M-\frac{3}{2}$	$\frac{3}{2}$	$3M-\frac{3}{2}$	-270	
	x_3	(1)	0	$\frac{3}{4}$	0	1	$\frac{3}{4}$	$\frac{1}{4}$	$-\frac{1}{4}$	15	
	x_2	(2)	0	$-\frac{1}{4}$	1	0	$-\frac{5}{4}$	$-\frac{3}{4}$	$\frac{3}{4}$	15	

So $x_1=0$, $x_2=15$, $x_3=15$, $\bar{x}_4=x_5=\bar{x}_6=0$

$(0,15,15,0,0,0)$ is optimal solution.

$z=90$

T2 minimize $z=3x_1+2x_2+4x_3$

s.t $2x_1+x_2+3x_3=60$

$3x_1+3x_2+5x_3\geq 120$

$x_1,x_2,x_3\geq 0$.

introduce the artificial variable $\bar{x}_4$, $\bar{x}_6$.

slack variable x_5

So the original LP problem can be transformed into

maximize $z=-3x_1-2x_2-4x_3-M(\bar{x}_4+\bar{x}_6)$.

s.t. $2x_1+x_2+3x_3+\bar{x}_4=60$

$3x_1+3x_2+5x_3-x_5+\bar{x}_6=120$

$x_1,x_2,x_3,\bar{x}_4,x_5,\bar{x}_6\geq 0$.

phase-1: maximize $-\bar{x}_4-\bar{x}_6$.

s.t. $2x_1+x_2+3x_3+\bar{x}_4=60$

$3x_1+3x_2+5x_3-x_5+\bar{x}_6=120$

$x_1,\cdots,\bar{x}_6\geq 0$.

transform that into the form:

Iteration	BV	Row	Coeifficient							RHS	Ratio
			z	x_1	x_2	x_3	$\bar{x}_4$	x_5	$\bar{x}_6$		
0	z	(0)	1	-5	-4	-8	0	1	0	-180	
	$\bar{x}_4$	(1)	0	2	1	3	1	0	0	60	20
	$\bar{x}_6$	(2)	0	3	3	5	0	-1	1	120	24
1	z	(0)	1	$\frac{1}{3}$	$-\frac{4}{3}$	0	$\frac{8}{3}$	1	0	-20	
	x_3	(1)	0	$\frac{2}{3}$	$\frac{1}{3}$	1	$\frac{1}{3}$	0	0	20	60
	$\bar{x}_6$	(2)	0	$-\frac{1}{3}$	$\frac{4}{3}$	0	$-\frac{5}{3}$	-1	1	20	15
2	z	(0)	1	-5	0	0	1	0	1	0	
	x_3	(1)	0	2	0	1	$\frac{3}{4}$	$\frac{1}{4}$	$-\frac{1}{4}$	15	
	x_2	(2)	0	-4	1	0	$-\frac{5}{4}$	$-\frac{3}{4}$	$\frac{3}{4}$	15	
3	z	(0)	1	0	0	$\frac{5}{2}$	$\frac{25}{8}$	$\frac{5}{8}$	$\frac{3}{8}$	$\frac{75}{2}$	
	x_1	(1)	0	1	0	$\frac{1}{2}$	$\frac{3}{8}$	$\frac{1}{8}$	$-\frac{1}{8}$	$\frac{15}{2}$	
	x_2	(2)	0	0	1	2	$\frac{1}{4}$	$-\frac{1}{4}$	$\frac{1}{4}$	45	
0'	z	(0)	1	3	2	4		$\frac{5}{8}$		$\frac{75}{2}$	
	x_1	(1)	0	1	0	$\frac{1}{2}$		$\frac{1}{8}$		$\frac{15}{2}$	
	x_2	(2)	0	0	1	2		$-\frac{1}{4}$		45	
1'	z	(0)	1	0	2	$\frac{5}{2}$		$\frac{1}{4}$		15	
	x_1	(1)	0	1	0	$\frac{1}{2}$		$\frac{1}{8}$		$\frac{15}{2}$	
	x_2	(2)	0	0	1	2		$-\frac{1}{4}$		45	
2'	z	(0)	1	0	0	$-\frac{3}{2}$		$\frac{3}{4}$		-75	
	x_1	(1)	0	1	0	$\frac{1}{2}$		$\frac{1}{8}$		$\frac{15}{2}$	15
	x_2	(2)	0	0	1	2		$-\frac{1}{4}$		45	$\frac{45}{2}$
3'	z	(0)	1	3	0	0		$\frac{9}{8}$		$-\frac{105}{2}$	
	x_3	(1)	0	2	0	1		$\frac{1}{4}$		15	
	x_2	(2)	0	-4	1	0		$-\frac{3}{4}$		15	

So $x_2 = 15$, $x_3 = 15$, $x_1 = 0 = x_5 = x_4 = x_6$.

$\langle 0, 15, 15, 0, 0, 0 \rangle$ is the optimal solution,

$z = 90$

T3. After introducting slack variables:

$$\begin{aligned} \text{maximize} \quad & z = 20x_1 + 6x_2 + 8x_3. \\ \text{s.t.} \quad & 8x_1 + 2x_2 + 3x_3 + x_4 = 200 \\ & 4x_1 + 3x_2 \qquad\qquad + x_5 = 100 \\ & 2x_1 \qquad + x_3 \qquad\qquad + x_6 = 50 \\ & \qquad\qquad x_3 \qquad\qquad\qquad + x_7 = 20 \\ & x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0. \end{aligned}$$

So the LP problem can be transformed into matrix type:

$$\begin{pmatrix} 1 & -c^T & 0 \\ 0 & A & I \end{pmatrix} \begin{pmatrix} z \\ x \\ x_s \end{pmatrix}$$

$$= \begin{pmatrix} 1 & -20 & -6 & -8 & 0 & 0 & 0 & 0 \\ 0 & 8 & 2 & 3 & 1 & 0 & 0 & 0 \\ 0 & 4 & 3 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} z \\ x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{pmatrix} = \begin{pmatrix} 0 \\ 200 \\ 100 \\ 50 \\ 20 \end{pmatrix}$$

After observing the simplex table, we can know x_1 row and x_2 row is $\langle 0,1,0,0,0 \rangle^T$ and $\langle 0,0,1,0,0 \rangle^T$.

the $x_B = \langle x_1, x_2, x_6, x_7 \rangle^T$, $x_N = \langle x_3, x_4, x_5 \rangle^T$

$$\because \quad A = \begin{pmatrix} 8 & 2 & 3 \\ 4 & 3 & 0 \\ 2 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \langle A \ \ I \rangle \begin{pmatrix} x \\ x_s \end{pmatrix} = \langle B \ \ N \rangle \begin{pmatrix} x_B \\ x_N \end{pmatrix} = \vec{b}$$

$$\therefore (B \quad N)\begin{pmatrix} x_1 \\ x_2 \\ x_6 \\ x_7 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 8 & 2 & 0 & 0 & 3 & 1 & 0 \\ 4 & 3 & 0 & 0 & 0 & 0 & 1 \\ 2 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 8 & 2 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} \frac{3}{16} & -\frac{1}{8} & 0 & 0 \\ -\frac{1}{4} & \frac{1}{2} & 0 & 0 \\ -\frac{3}{8} & \frac{1}{4} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad C_B^T = (c_1 \quad c_2 \quad c_3 \quad c_4).$$

$$y^* = C_B^T B^{-1} = (\tfrac{3}{16}c_1 - \tfrac{1}{4}c_2 - \tfrac{3}{8}c_3, -\tfrac{1}{8}c_1 + \tfrac{1}{2}c_2 + \tfrac{1}{4}c_3, c_3, c_4)$$

$$= (\tfrac{9}{4}, \tfrac{1}{2}, 0, 0) \Rightarrow \begin{cases} c_1 = 20 \\ c_2 = 6 \\ c_3 = 0 \\ c_4 = 0 \end{cases} \Rightarrow C_B^T = (20 \quad 6 \quad 0 \quad 0)$$

$$\therefore y^* A - c^T = (\tfrac{9}{4}, \tfrac{1}{2}, 0, 0)\begin{pmatrix} 8 & 2 & 3 \\ 4 & 3 & 0 \\ 2 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} - (20, 6, 8)$$

$$= (20, 6, \tfrac{27}{4}) - (20, 6, 8) = (0, 0, -\tfrac{5}{4})$$

$$B^{-1}A = \begin{pmatrix} \frac{3}{16} & -\frac{1}{8} & 0 & 0 \\ -\frac{1}{4} & \frac{1}{2} & 0 & 0 \\ -\frac{3}{8} & \frac{1}{4} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 8 & 2 & 3 \\ 4 & 3 & 0 \\ 2 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & \frac{9}{16} \\ 0 & 1 & -\frac{3}{4} \\ 0 & 0 & -\frac{1}{8} \\ 0 & 0 & 1 \end{pmatrix}$$

$$y^* b = (\tfrac{9}{4}, \tfrac{1}{2}, 0, 0)\begin{pmatrix} 200 \\ 100 \\ 50 \\ 20 \end{pmatrix} = 450 + 50 = 500$$

$$B^{-1}b = \begin{pmatrix} \frac{3}{16} & -\frac{1}{8} & 0 & 0 \\ -\frac{1}{4} & \frac{1}{2} & 0 & 0 \\ -\frac{3}{8} & \frac{1}{4} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 200 \\ 100 \\ 50 \\ 20 \end{pmatrix} = \begin{pmatrix} 25 \\ 0 \\ -50 \\ 20 \end{pmatrix}$$

∴ The completed form is:

Iteration	BV	row	Coefficient								RHS.
			z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	
0	z	(0)	1	0	0	$-\frac{5}{4}$	$\frac{9}{4}$	$\frac{1}{2}$	0	0	500
	x_1	(1)	0	1	0	$\frac{9}{16}$	$\frac{3}{16}$	$-\frac{1}{8}$	0	0	25
	x_2	(2)	0	0	1	$-\frac{3}{4}$	$-\frac{1}{4}$	$\frac{1}{2}$	0	0	0
	x_6	(3)	0	0	0	$-\frac{1}{8}$	$-\frac{3}{8}$	$\frac{1}{4}$	1	0	-50
	x_7	(4)	0	0	0	1	0	0	0	1	20